

Overview of metabolism in the fed and fasting state

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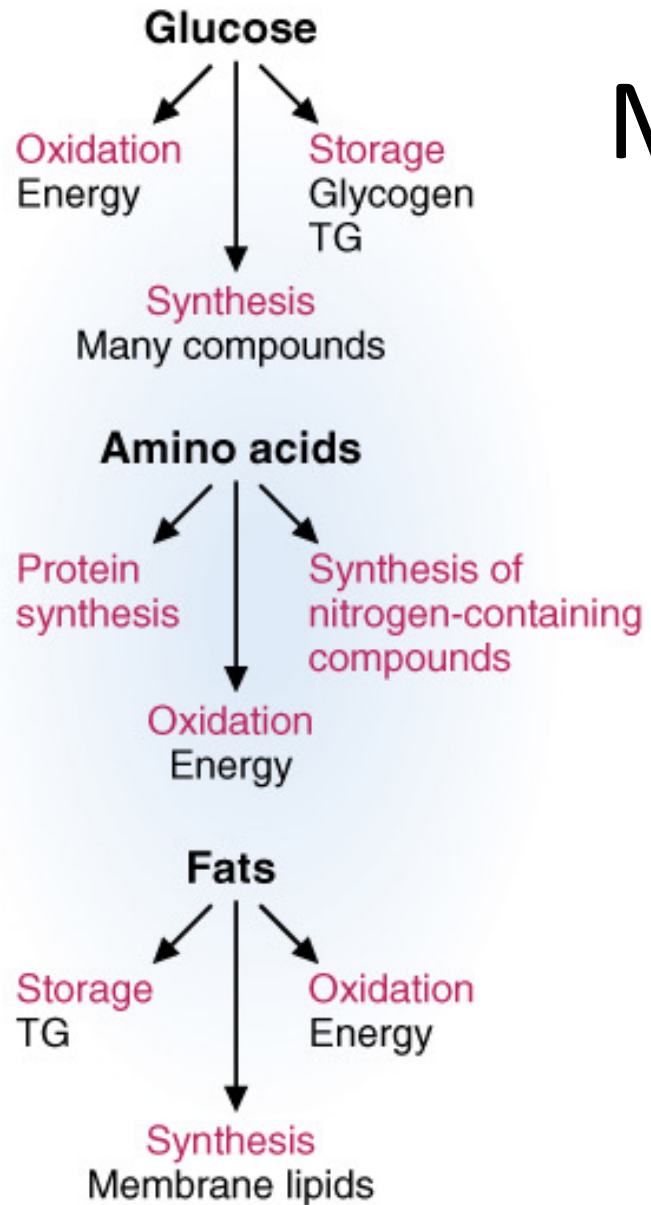
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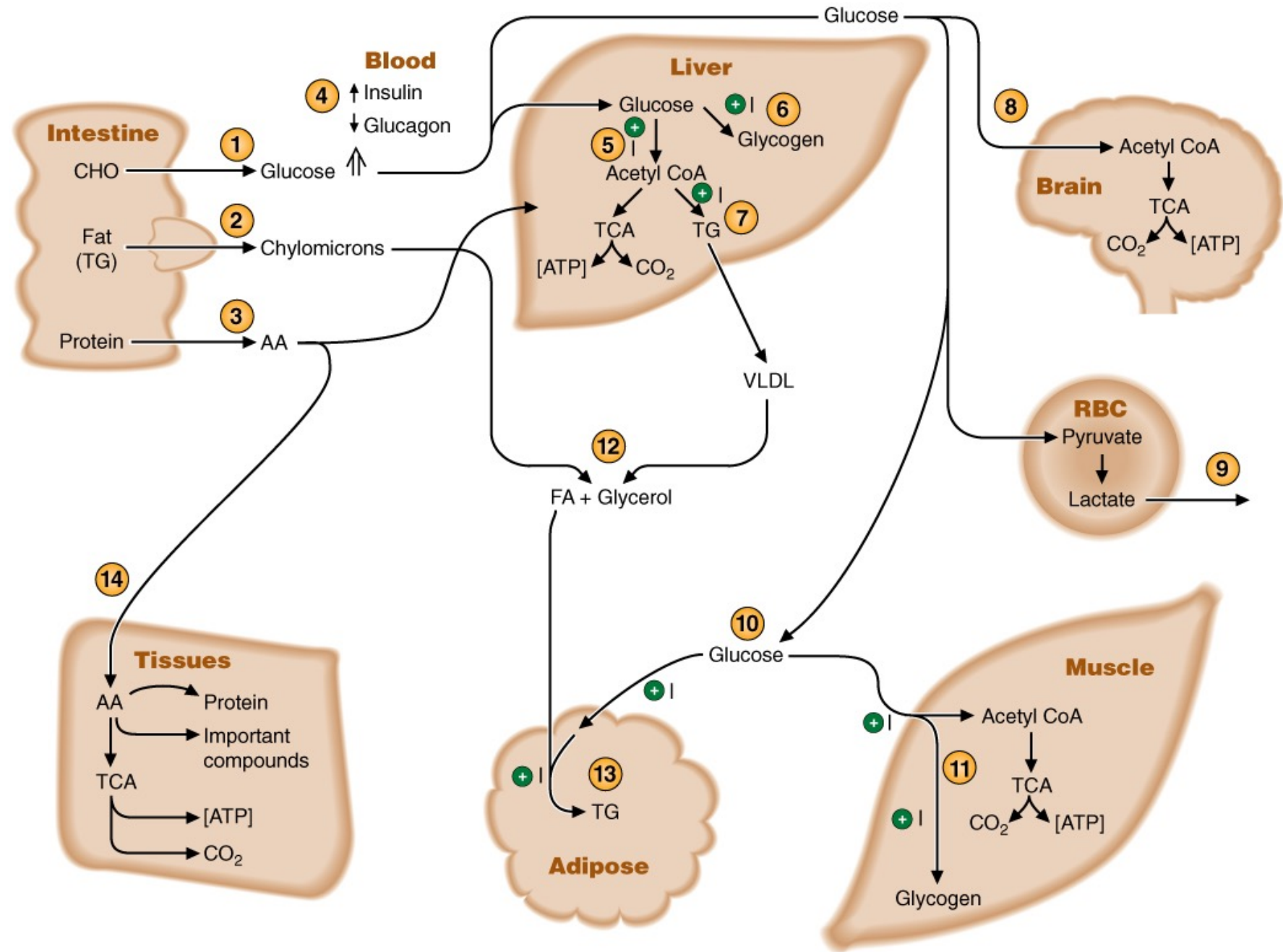
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This DLA has two videos and provides an overview of metabolism in the fed and fasting state

Major fates of fuel in fed state



Metabolism in **FED** State



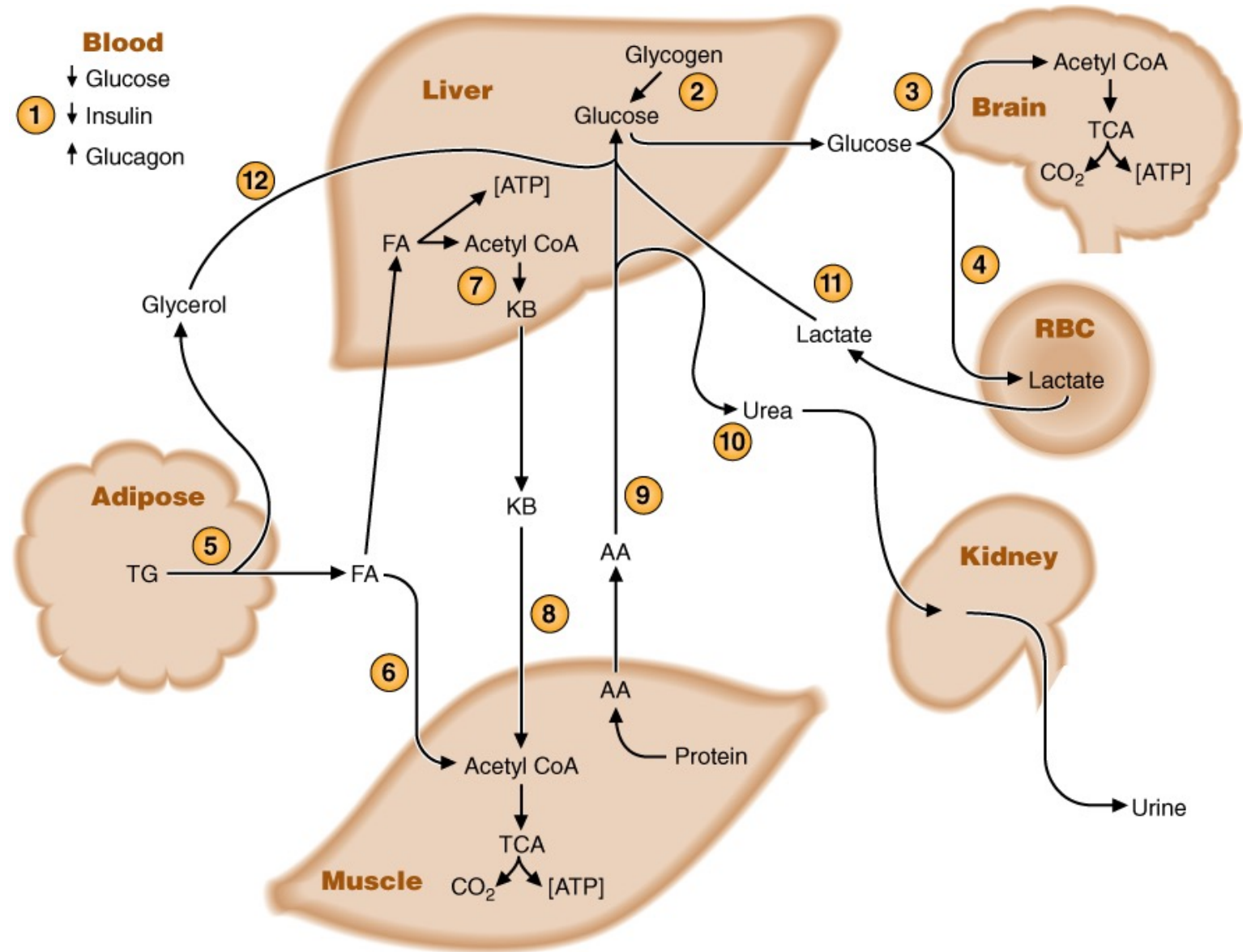
Metabolism in **FED** state

1. **Absorption** of **dietary glucose** increases plasma glucose
2. Dietary **TAGs** are **absorbed** and then **transported via chylomicrons**
3. Dietary **protein** is **absorbed as amino** acids
4. **Increased** plasma **glucose** = **results in Insulin secretion** and decreases glucagon secretion (**Increases Insulin:Glucagon ratio**)
5. **Liver Glycolysis** is stimulated
6. **Liver Glycogenesis** is favored
7. **FA Synthesis from** the **excess Acetyl-CoA** is stimulated & **TAG** formation is stimulated
8. **Brain uses** plasma **glucose** as an energy source (**complete oxidation** = **Glucose** used all the way **to CO₂ and H₂O**)
9. **RBCs** uses **Glucose** as an energy source **to** form **Lactate** which is **then** transported via **Cori cycle** to the **Liver**
10. **Glucose** entry **into Adipocytes** is favored due to presence of **GLUT-4 on membrane via Insulin**, to be **stored as Fat**
11. **Glucose** entry **into Muscle** is favored due to presence of **GLUT-4 on membrane via Insulin** and **Glycogen synthesis** is stimulated
12. **Lipoprotein-Lipase** in **endothelium** of blood vessels of **Adipocytes & Muscle** is active, **forms FAs & Glycerol**, **taken up by Adipocytes**
13. **In Adipocytes**, **formation** and **storage** of **TAG** is favored. **Glycerol-phosphate** is **provided by DHAP** from **Glycolysis**.
14. **Amino** acid **entry** and uptake **into all tissues** is favored. **Protein synthesis** is **active**

Metabolism
10- 12 hours
after meal

FASTED

state
(Overnight fast)



Metabolism 10- 12 hours after meal, **FASTED** state (Overnight fast)

1. A **fall in** plasma **glucose** **reduces** the **Insulin:Glucagon ratio**
2. **Liver Glycogenolysis** is stimulated
3. Plasma **Glucose** is **exclusively for brain** metabolism (**complete oxidation = Glucose used all the way to CO₂ and H₂O**)
4. **RBCs** use glucose and **form Lactate**.
5. **Adipose** tissue **Lipolysis** is active (**TAG** broken down **to Free fatty acids & Glycerol** by HSL)
6. **FAs** are used **for energy** by **Skm & Cardiac** muscle
7. **β-oxidation** in the **Liver** provides **energy for** the **Liver** & **forms Ketone Bodies**
8. **Ketone Bodies** are **used by** the **Muscles** for energy
9. **Muscle Proteolysis** is active and **releases amino** acids, **sent to Liver** and used **for Gluconeogenesis**.
10. **Ammonia** generated **by amino acid metabolism** is **detoxified to Urea** which is **lost through** the **kidneys (urine)**
11. **Cori cycle** transports **Lactate to Liver** which is used **for Gluconeogenesis**
12. **Glycerol** formed **in Adipose** tissue is also **used as a contributing precursor for Gluconeogenesis**